

AVA CHEN

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ACADEMIC POSITIONS

Stanford University

Postdoctoral Scholar in Mechanical Engineering, Advisor: Allison Okamura

2025 – present

EDUCATION

Columbia University

Ph.D in Mechanical Engineering, Advisor: Matei Ciocarlie

Thesis: “*Robotic Exoskeletons to Assist Volitional Motor Control of the Paretic Hand*”

2019 – 2025

New York, NY

Columbia University

M.S. in Mechanical Engineering

2019 – 2021

New York, NY

Massachusetts Institute of Technology (MIT)

B.S. in Mechanical Engineering, Advisors: Hugh Herr, Arthur Petron

Thesis: “*Effectiveness of Active Cooling on Torque Performance for Prosthetic Applications*”

2013 – 2017

Cambridge, MA

HONORS

Wu Tsai Human Performance Alliance Research Fellowship

2026 – present

NIH Ruth L. Kirschstein National Research Service Award (NRSA) F31 – NICHD

2023 – 2025

Robotics: Science & Systems (RSS) Pioneer

2024

Rising Star in ME 2022 at Stanford University

2022

PUBLICATIONS

Peer-Reviewed Journal Articles

[* indicates equal contribution]

- [J.5] J. Xu*, R. Wang*, S. Shang*, **A. Chen**, L. Winterbottom, L. Hsu, W. Chen, K. Ahmed, P. L. La Rotta, X. Zhu, D. M. Nilsen, J. Stein, and M. Ciocarlie, “ChatEMG: Synthetic Data Generation to Control a Robotic Hand Orthosis for Stroke.” *IEEE Robotics and Automation Letters*, 10, 2, 907-914. (2025)
 - Presented in *2025 IEEE Intl. Conference on Robotics and Automation (ICRA)*.
- [J.4] L. Winterbottom*, **A. Chen***, R. Mendonca, D.M. Nilsen, M. Ciocarlie, and J. Stein. “Clinician perceptions of a novel wearable robotic hand orthosis for post-stroke hemiparesis.” *Disability and Rehabilitation*, 47, 6, 1577–1586. (2024)
- [J.3] J. Palacios*, A. Deli-Ivanov*, **A. Chen***, L. Winterbottom, D. M. Nilsen, J. Stein, and M. Ciocarlie, “Grasp Force Assistance via Throttle-based Wrist Angle Control on a Robotic Hand Orthosis for C6-C7 Spinal Cord Injury.” *IEEE Transactions on Medical Robotics and Bionics*, 7, 1, 149-155. (2024)
 - Presented in *2024 IEEE RAS/EMBS Intl. Conference on Biomedical Robotics and Biomechatronics (BioRob)*.
- [J.2] **A. Chen**, L. Winterbottom, S. Park, J. Xu, D. M. Nilsen, J. Stein, and M. Ciocarlie, “Thumb Stabilization and Assistance in a Robotic Hand Orthosis for Post-Stroke Hemiparesis.” *IEEE Robotics and Automation Letters*, 7, 3, 8276-8282. (2022)
 - Presented in *2022 IEEE RAS/EMBS Intl. Conference on Biomedical Robotics and Biomechatronics (BioRob)*.
Finalist, BioRob2022 Best Paper Award
- [J.1] **A. Chen**, K. Kim, and P.S. Shamble. “Rapid mid-jump production of high-performance silk by jumping spiders.” *Current Biology*, 31, R1422-R1423. (2021)

Peer-Reviewed Conference Papers

- [C.8] **A. Chen**, M. C. Coram, C. du Pasquier, and A. M. Okamura. “Miniaturized Pneumatic Actuator Array for Multipoint Deep Pressure Tactile Stimulation.” Work-in-Progress Paper in *2026 IEEE Haptics Symposium*.
- [C.7] J. Xu*, **A. Chen***, L. Winterbottom, J. Palacios, P. Chivukula, D. M. Nilsen, J. Stein, and M. Ciocarlie. “Reciprocal Learning of Intent Inference with Augmented Visual Feedback for Stroke.” In *2025 IEEE Intl. Conference on Rehabilitation Robotics (ICORR)*.
- [C.6] K. Lee, R. Wang, **A. Chen**, L. Winterbottom, H. M. C. Leung, L. M. DiSalvo, I. Xu, J. Xu, D. M. Nilsen, J. Stein, X. Zhou, and M. Ciocarlie. “Fabric Sensing of Intrinsic Hand Muscle Activity.” In *2025 IEEE Intl. Conference on Rehabilitation Robotics (ICORR)*.

- [C.5] **A. Chen***, K. Lee*, L. Winterbottom, J. Xu, C. Lee, G. Munger, A. Deli-Ivanov, D. M. Nilsen, J. Stein, and M. Ciocarlie, “Volitional Control of the Paretic Hand Post-Stroke Increases Finger Stiffness and Resistance to Robot-Assisted Movement.” In *2024 IEEE RAS/EMBS Intl. Conference on Biomedical Robotics and Biomechatronics (BioRob)*.
- [C.4] P. L. La Rotta*, J. Xu*, **A. Chen**, L. Winterbottom, W. Chen, D. M. Nilsen, J. Stein, and M. Ciocarlie, “Meta-Learning for Fast Adaptation in Intent Inference on a Robotic Hand Orthosis for Stroke.” In *2024 IEEE/RSJ Intl. Conference on Intelligent Robots and Systems (IROS)*.
- [C.3] **A. Chen**, L. Winterbottom, K. O’Reilly, S. Park, D. M. Nilsen, J. Stein, and M. Ciocarlie. “Design of Spiral-Cable Forearm Exoskeleton to Provide Supination Adjustment for Hemiparetic Stroke Subjects.” In *2022 IEEE Intl. Conference on Rehabilitation Robotics (ICORR)*.
- [C.2] J. Xu, C. Meeker, **A. Chen**, L. Winterbottom, M. Fraser, S. Park, L. M. Weber, M. Miya, D. M. Nilsen, J. Stein, and M. Ciocarlie. “Adaptive Semi-Supervised Intent Inference to Control a Powered Hand Orthosis for Stroke.” In *2022 IEEE Intl. Conference on Robotics and Automation (ICRA)*.
- [C.1] T. Cervantes, W.E. Byun*, **A. Chen***, K. Kim*, K. Nealon*, J. Connor, and A. Slocum. “A Device for Quantitative Analysis of the Thumb Ulnar Collateral Ligament.” ASME. *Frontiers in Biomedical Devices, 2018 ASME Design of Medical Devices Conference*.

Submitted for Publication

- [S.1] R. Wang*, X. Zhu*, **A. Chen**, J. Xu, L. Winterbottom, D. M. Nilsen, J. Stein, and M. Ciocarlie. “ReactEMG: Stable, Low-Latency Intent Detection from sEMG via Masked Modeling” (2025, under review)

Workshop and Symposium Contributions

- [W.8] L. Winterbottom, **A. Chen**, D. M. Nilsen, R. Mendonca, J. Xu, K. Lee, M. Ciocarlie, and J. Stein. “Motor learning techniques to enhance training with robotic hand orthoses for stroke survivors: challenges and opportunities.” In *American Occupational Therapy Association (AOTA) INSPIRE 2025*.
- [W.7] **A. Chen**, J. Xu, K. Lee, L. Winterbottom, D. M. Nilsen, J. Stein, and M. Ciocarlie. “Bidirectional Human-Robot Feedback and Physical Effects of Assisted Manipulation with a Robotic Hand Orthosis for Stroke.” In *New England Manipulation Symposium (NEMS) 2024*.
- [W.6] **A. Chen**. “Assisting Impaired Dexterity with User-Driven Robotic Hand Exoskeletons.” In *Robotics: Science and Systems (RSS) workshop: RSS Pioneers 2024*.
- [W.5] K. Lee, L. M. DiSalvo, I. Xu, **A. Chen**, X. Zhou, and M. Ciocarlie. “Fabric EMG Sensing for Robotic Orthosis Control.” In *2024 IEEE Intl. Conference on Biomedical Robotics and Biomechatronics (BioRob) workshop: Building Responsive Body-Machine Interfaces with Biosignals and Robotic Exoskeletons*.
- [W.4] L. Winterbottom, D. Nilsen, R. Mendonca, **A. Chen**, S. Lin, K. Carroll, J. Xu, M. Ciocarlie, and J. Stein. “Collaboration between Occupational Therapists, Engineers, and People with Neurological Conditions in the Development of Wearable Robotic Devices.” In *American Occupational Therapy Association (AOTA) INSPIRE 2024*.
- [W.3] J. Palacios*, A. Deli-Ivanov*, **A. Chen**, L. Winterbottom, D. M. Nilsen, J. Stein, and M. Ciocarlie. “Towards Tenodesis-Modulated Control of an Assistive Hand Exoskeleton for SCI.” In *2023 IEEE/RSJ Intl. Conf. on Intelligent Robots and Systems (IROS) workshop: Assistive Robotics for Citizens*.
- [W.2] L. Winterbottom, K. Carroll, S. Lin, **A. Chen**, R. Mendonca, D. M. Nilsen, M. Ciocarlie, and J. Stein. “Stroke Survivors’ Perspectives on the Design of a Novel Wearable Robotic Hand Brace.” In *2022 Janet Falk-Kessler Distinguished Lectureship and Day of Scholarship*.
- [W.1] L. Winterbottom, D. Nilsen, R. Mendonca, **A. Chen**, J. Xu, M. Ciocarlie, and J. Stein. “Perspectives of Individuals with C6-C7 Spinal Cord Injury on the Design of a Novel Robotic Hand Brace.” In *2021 Janet Falk-Kessler Distinguished Lectureship and Day of Scholarship*.

Patents

- [P.1] M. Ciocarlie, J. Stein, **A. Chen**, S. Park, D. M. Nilsen. “Robotic Hand Orthosis For Stroke”, Application #: US 63/249,456

INDEPENDENT FUNDING

<i>On-field wearable haptic training to reduce knee injury risk during running cuts.</i>				
Stanford Wu Tsai Human Performance Research Fellowship	9/2026–8/2027	\$87,000 + benefits	PI: Chen	
<i>Impact of biofeedback and task-specific training with a robotic hand orthosis on voluntary muscle modulation for rehabilitation post-stroke.</i>				
NIH F31 1F31HD111301	NICHD (NCMRR)	8/2023–1/2025	\$72,587	PI: Chen

INVITED TALKS

Interpreting Human Action and Intent with Wearable Robots	June 2026
Stanford AI4ALL Guest Lecture	
Enhancing Movement and Sensation with Haptic Garments	Mar. 2026
IEEE Haptics Symposium 2026, Cross-Cutting Challenge: Touch, Embodiment, and Multisensory Future	
Assisting Human Autonomy with Wearable Robots	Dec. 2025
Duke University, Duke Robotics Seminar	
Lessons in Dexterity From Observing Spiders, Humans, and Robots	Dec. 2025
Duke University, Robot Dexterity Lab (DexLab) Meeting	
Robots to Assist and Observe Biological Autonomy, Behavior, and Interaction	Dec. 2025
UT Austin, MERGe Lab Meeting	
Robotic Hand Exoskeletons to Assist and Rehabilitate Impaired Dexterity	June 2024
Harvard University, Harvard Biorobotics Lab Meeting	
Robotic Hand Orthoses for Assistance and Rehabilitation After Stroke (Co-Speaker)	Sept. 2023
Global Perspectives on Medicine, Rehabilitation and Robotics Webinar Series	
MyHand: a Wearable Hand Orthosis for Stroke (Co-Speaker)	Oct. 2021
IROS Workshop, Challenges and Opportunities of Human-Robot Symbiosis: from Wearable Robots to Neurorobotics	
How Jumping Spiders Use Silk to Orient Themselves in Midair	Oct. 2018
Harvard University, Harvard Bauer Forum	
How Jumping Spiders Jump	Oct. 2018
Broad Institute, CEE 35th Anniversary Celebration	

TEACHING EXPERIENCE AND MENTORSHIP

University Courses	
Teaching Assistant, Columbia MECE E4602 – Introduction to Robotics	Fall 2020
Lab Assistant, Harvard LS50 – Integrated Science	Spring 2018, Spring 2019
Pedagogical Training	
Participant, Columbia Center for Teaching and Learning (CTL) Teaching Development Program	2022 – 2025
Talks on Teaching	
Columbia CTL “Wowza!” CIRTLL Discussion Series – Speaker, “Supporting Teaching as Scholarship”	Mar. 2024
Columbia CTLGrads Journal Club workshop – Speaker, “Effective Teaching Online, Real-Time”	Oct. 2023
Extracurricular	
Columbia Engineering Your PhD – Invited Panelist, “Insights from Experienced TAs”	Aug. 2023, Aug. 2024
Academic Mentor, Women in Science at Columbia (WISC)	2020, 2021, 2023
Research Mentor, Columbia University Engineering the Next Generation (ENG)	Summer 2022
Research Mentor and Teaching Assistant, Research Science Institute (RSI)	Summer 2014

RESEARCH STUDENTS SUPERVISED

Sarah Glomski , Stanford Ph.D Mechanical Engineering	2025 – present
Sunny Li , Stanford Ph.D Mechanical Engineering	2025 – present
Sherry Choi , Stanford Undergraduate Visiting Intern	Summer 2026
Jasmine Yang , Stanford SURF Fellow	Summer 2026
Cecilia Wu , Stanford M.S. Computer Science	Fall 2025
Andrew Chin , Columbia M.S. Robotics	2024 – 2025
Elise Yang , Columbia Undergraduate	2024 – 2025

Shiyao Marcus Lam , Columbia Undergraduate	2024 – 2025
Grace Munger , Columbia Undergraduate [C.5]	2023 – 2025
Matheu Campbell , Columbia Undergraduate	2023 – 2024
Akshay Venkatesan , Columbia M.S. Data Science	2023 – 2024
Runsheng Wang , Columbia M.A. Statistics [J.5, C.6, S.1, W.8]	2023 – 2024
Connor Lee , Columbia Undergraduate [C.5]	2023 – 2024
Alexandra Deli-Ivanov , Columbia Undergraduate [J.3, C.5, W.3]	2022 – 2024
Joaquin Palacios , Columbia Undergraduate and M.S. Robotics [J.3, C.7, W.3]	2021 – 2024
Pedro Leandro La Rotta , Columbia M.S. Robotics [C.4, J.5]	2023
Katherine O'Reilly , Columbia Undergraduate [C.3]	2020 – 2023
Carolyn David , Columbia M.S. Biomedical Engineering	2022 – 2023
Preethika Chivukula , Columbia M.S. Biomedical Engineering [C.7]	2021 – 2022
Ashley Reyes , Columbia ENG Student	Summer 2022
Brayan Ramos , Columbia ENG Student	Summer 2022
Ciara Little , Columbia Undergraduate	2020 – 2021
Katelyn G. Mitchell , Columbia Undergraduate	2020 – 2021
Frederick Horne , Harvard Undergraduate	2019
Rowen VonPlagenhoef , Harvard Undergraduate	2019
Eliot Burnes , Harvard Undergraduate	2018 – 2019
Henry Burnes , Harvard Undergraduate	2018 – 2019
Lincoln Sorscher , Harvard Undergraduate	2018
Cheng Lu , RSI Scholar	Summer 2014

SERVICE

University and Conference Service

Program Committee, RSS Pioneers 2025	2025
Workshop Co-Organizer, BioRob 2024	2024
“ Building Responsive Body-Machine Interfaces with Biosignals and Robotic Exoskeletons ”	
CIRTL Fellow, Columbia University Center for Teaching and Learning	2023 – 2024
Conference Volunteer, Robotics: Science and Systems (RSS)	2022

External Paper Reviewer

IEEE Transactions on Haptics (ToH)	2025, 2026
IEEE/RSJ Intl. Conference on Intelligent Robots and Systems (IROS)	2024, 2026
Eurohaptics	2026
IEEE RAS/EMBS Intl. Conference on Biomedical Robotics & Biomechatronics (BioRob)	2022, 2024, 2026
IEEE Robotics and Automation Letters (RA-L)	2021, 2022, 2025
Robosoft	2025
Disability and Rehabilitation	2025
IEEE Intl. Conference on Rehabilitation Robotics (ICORR)	2022, 2025
IEEE Transactions on Robotics (TRO)	2024
IEEE Intl. Conference on Robotics and Automation (ICRA)	2021, 2022, 2024
IEEE Transactions on Medical Robotics and Bionics (T-MRB)	2023
Scientific Reports	2022, 2023
IEEE Intl. Conference on Robot and Human Interactive Communication (RO-MAN)	2022, 2023
IEEE Transactions on Neural Systems and Rehabilitation Engineering (TNSRE)	2020

Science Volunteering and Outreach

Volunteer, Stanford Institutes of Medicine Summer Research Program	2025
Question Reviewer, U.S. Dept. of Energy National Science Bowl (NSB)	2023, 2024
Volunteer / Paper Reviewer, Research Science Institute (RSI) at MIT	2015, 2018, 2024, 2025
Columbia Engineering Achievers in Graduate Education (EngAGE) – Invited Panelist	Mar. 2024
Columbia WISC STEM Field Exploration Fair – Invited Panelist, “Behind the Lab Scenes”	Apr. 2022
Judge, Kentucky Science and Engineering Fair	2021
Judge, MIT Mechanical Engineering Research Exhibition	2020
Question Writer, USA Biolympiad (USABO)	2019
Volunteer, Adaptive Climbing Group NY	2019
Judge, Sweden Research Academy for Young Scientists (RAYS)	2015

PREVIOUS POSITIONS

Harvard Dept. of Organismic & Evolutionary Biology, Shamble Lab Research Assistant with Paul Shamble <i>Studied jumping dynamics and mid-air silk production of jumping spiders (<i>S. scenicus</i>) [J.1]</i>	2017 – 2019
Dephy, Inc. Mechanical Engineering Intern <i>Designed and built validation equipment for full-system stress testing</i>	Summer 2017, Fall 2018
MIT Media Lab, Biomechatronics Group Undergraduate Researcher with Hugh Herr, Arthur Petron, and Matt Carney <i>Worked on FitSocket project for soft-tissue characterization and on active motor cooling</i>	2013 – 2017
Apple Inc. Product Design Validation Engineer Intern <i>Worked on design, usability, and validation for mechanical features in hardware products</i>	Summer 2016
Formlabs Mechanical Engineering Intern <i>Electromechanical design for early versions of Form 2 and Form Cure products</i>	Summer 2015